

Day 5 Evidence Workbooklet — Instructor Key

Boolean Combinations and Week 1 Synthesis

Engineering practice → separate evidence → Boolean expression → go/no-go reasoning → support route

Instructor key. Same layout as student copy; solutions filled in response boxes. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	____/40		Mastery check		secure / developing / needs support

The pump-flow team now has three kinds of evidence from this week: current stored values from Day 2, typed and converted values from Day 3, and boundary comparisons from Day 4. Today the team combines separate evidence checks into a simple go/no-go readiness expression.

The problem today is Boolean combination, not branch routing. You will predict whether combined conditions are **True** or **False**, explain which part controls the result, and name whether your support need is state, type, boundary, or Boolean reasoning.

Engineering task	Decide whether a pump record is ready for supervisor review using separate evidence checks that must all be traceable.
Computational move	Combine condition results with and , or , and not while preserving the reason each condition is true or false.
Python focus	Boolean values, comparison results, and , or , not , parentheses, and compound expression explanation.
Evidence to keep	typed value, boundary result, label check, combined truth value, controlling condition, and support category.

Day 5 Boolean rules

1. A **and** B is **True** only when both parts are **True**.
2. A **or** B is **True** when at least one part is **True**.
3. **not** A reverses a Boolean result.
4. Parentheses make the intended grouping visible.
5. A reliable go/no-go note names the separate evidence checks before naming the combined result.

1 Read separate evidence before combining it

[recognize](#)[predict](#)

Use the week-one pump record below. Each named condition stores one separate evidence result.

```
1 # Typed values and boundary limits
2 flow_lpm = 20.1
3 low_limit_lpm = 19.6
4 high_limit_lpm = 20.4
5 approval_label = "reviewed"
6 operator_initials = "AM"
7 retry_count = 0
8 sensor_ready = True
9
10 # Separate evidence checks
11 lower_ok = flow_lpm >= low_limit_lpm
12 upper_ok = flow_lpm <= high_limit_lpm
13 flow_ok = lower_ok and upper_ok
14 approval_ok = approval_label == "reviewed"
15 setup_ok = sensor_ready and operator_initials != ""
16 no_retry_needed = retry_count == 0
17 record_ready = flow_ok and approval_ok and setup_ok and no_retry_needed
```

Pts.	Prompt	My evidence
1	Which two comparisons create the separate lower and upper boundary checks?	Key: flow_lpm >= low_limit_lpm and flow_lpm <= high_limit_lpm.
1	Which condition checks the text label rather than numeric flow?	Key: approval_ok = approval_label == "reviewed".
1	Which condition checks whether a retry is needed?	Key: no_retry_needed = retry_count == 0.
1	Which final expression combines all readiness evidence?	Key: record_ready = flow_ok and approval_ok and setup_ok and no_retry_needed.

2 Trace an and combination from separate evidence

[trace](#) [explain](#)

Complete the table. Do not skip directly to the final result; show each part.

Pts.	Combined condition	Left part	Right part	Combined result and reason
2	lower_ok and upper_ok	Key: True	Key: True	Key: Both boundary checks pass, so flow_ok is True.
2	sensor_ready and operator_initials != ""	Key: True	Key: True	Key: Sensor ready and initials present, so setup passes.
2	flow_ok and approval_ok	Key: True	Key: True	Key: Both flow and approval checks pass.
2	setup_ok and no_retry_needed	Key: True	Key: True	Key: Setup passes and no retry is needed.

State/type/boundary connection

A Boolean expression is only as trustworthy as the evidence pieces inside it. If a part came from a string label, a converted number, or a boundary comparison, keep that source visible in your reason.

3 Use or and not for follow-up evidence

[predict](#)[explain](#)

A record may need follow-up if at least one risk is present. Complete the table and explain which part controls the result.

```
1 needs_flow_review = not flow_ok
2 needs_approval_review = not approval_ok
3 needs_setup_review = not setup_ok
4 needs_followup = needs_flow_review or needs_approval_review or needs_setup_review
```

Pts.	Expression	True or False?	Reason
2	not flow_ok	Key: False	Key: flow_ok is True, so its negation is False.
2	not approval_ok	Key: False	Key: approval_ok is True.
2	needs_flow_review or needs_approval_review	Key: False	Key: Both review flags are false.
2	needs_followup	Key: False	Key: No listed review flag is true in the baseline.

Common trap to check

or does not mean every risk is present. It means at least one part is True. A good note says which part made the expression true.

4 Build a Week 1 condition table

[trace](#) [transfer](#)

For each case, use Day 3 type/label evidence and Day 4 boundary evidence before deciding the combined result.

Pts.	Flow	Approval label	flow_ok	approval_ok	setup_ok	record_ready and reason
2	20.1	"reviewed"	Key: True	Key: True	Key: True	Key: True; all required evidence checks pass.
2	20.5	"reviewed"	Key: False	Key: True	Key: True	Key: False; flow is above the upper limit.
2	20.0	"pending"	Key: True	Key: False	Key: True	Key: False; approval label is not reviewed.
2	19.6	"reviewed" with blank initials	Key: True	Key: True	Key: False	Key: False; blank initials fail setup even though flow is at the included lower boundary.

Bridge to Week 2

Week 2 will use these Boolean results to choose routes with **if/else**. Day 5 stops one step earlier: decide the truth value and explain the evidence, but do not write a branch yet.

5 Debug a Boolean readiness claim

[debug](#) [test](#) [explain](#)

A teammate writes this note after seeing that `approval_label` stores a non-empty string:

Readiness note to debug

The record is ready because the flow is in range or the approval label has text.

Pts.	Prompt	My response
2	What Day 3 mistake appears in the teammate's note?	Key: It treats any non-empty approval text as enough; the label must be checked explicitly.
2	What Day 4 boundary evidence is still needed before trusting <code>flow_ok</code> ?	Key: Both lower and upper inclusive comparisons must be shown: <code>flow_lpm >= low_limit_lpm</code> and <code>flow_lpm <= high_limit_lpm</code> .
2	Why is <code>or</code> unsafe if the engineering rule requires both flow evidence and approval evidence?	Key: <code>or</code> can pass when only one condition is true; readiness requires all required evidence checks.
2	Rewrite the readiness note so it names the separate evidence checks and uses <code>and</code> correctly.	Key: The record is ready only if <code>flow_ok</code> , <code>approval_ok</code> , <code>setup_ok</code> , and <code>no_retry_needed</code> are all true. In this base-line case, all are true.

Week 1 synthesis habit

When a combined expression surprises you, diagnose the support need by source: current state from Day 2, type/cast/truthiness from Day 3, boundary comparison from Day 4, or Boolean connector from Day 5.

6 Mastery check and support route

[transfer](#)[explain](#)

Use your own evidence from this workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	Give one example where an and expression is false because one required evidence check failed.	Key: If flow is 20.5, upper_ok is false, so flow_ok and then record_ready are false.
2	In one or two sentences, explain how Day 2 state, Day 3 type evidence, and Day 4 boundary evidence work together in Day 5.	Key: Day 2 identifies current values, Day 3 makes sure values have safe types/labels, and Day 4 creates boundary results. Day 5 combines those evidence pieces with Boolean connectors.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can combine separate evidence checks with and , or , and not , and explain which part controls the result.
☐	Developing: I can evaluate some combined conditions, but I still need support identifying which evidence check failed.
☐	Needs support: I need a smaller example before I can combine state, type, boundary, and Boolean evidence.

My next support move

My issue is mostly: setup / Python syntax / tracing / state history / type conversion / boundary cases / Boolean connectors / confidence / time.

The first evidence source where I got uncertain was: _____

My next small action is: ask a partner / ask instructor / rebuild the condition table / test one failed condition at a time / try the recovery version.

Workbooklet target: I can combine typed values, boundary checks, approval evidence, and Boolean connectors into a traceable Week 1 go/no-go explanation.

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